

No. of Functional Features Included in the Solution

Date	23 June 2026
Team ID	SWTID-2026-1377
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features implemented in the **Voice-Based Concept Understanding Analyser**. Each feature contributes to evaluating students' conceptual understanding through voice or text explanations using Artificial Intelligence and Natural Language Processing (NLP) techniques..

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Home Page	Introduces the Voice-Based Concept Understanding Analyser, its objectives, and provides navigation to other application pages.	Frontend (UI)	Done	0.5
2	About Page	Explains the project, workflow, technologies used, and benefits of AI-based concept evaluation.	Frontend (UI)	Done	0.5
3	Crop Prediction Form	Allows students to record voice explanations or manually enter text responses for concept evaluation.	Frontend (UI)	Done	0.75
4	Data Preprocessing Pipeline	Converts spoken explanations into text using the browser's Web Speech API before analysis.	Backend / Speech Processing	Done	0.75
5	ML Model Training	Performs text preprocessing, semantic similarity analysis, concept coverage evaluation, and misconception detection using NLP techniques	Backend / NLP Engine	Done	0.75
6	Model Evaluation & Selection	Calculates mastery scores, identifies missing concepts, and generates personalized learning feedback.	Backend / Evaluation Engine	Done	0.75
7	Real-Time Crop Prediction	Processes user responses through the Flask backend and instantly displays concept analysis results.	Backend / Flask Integration	Done	0.75
8	Application Testing	End-to-end testing of voice recording, concept analysis, feedback generation, and report display to ensure correct functionality.	Testing / Validation	Done	0.25

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (Voice & Text Input, Speech-to-Text Conversion, NLP Concept Analysis, Mastery Score Generation, Real-Time Concept Evaluation, Feedback Generation)
Additional / Nice-to-Have Features	2 (Home Page, About Page, Application Testing)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home Page, About Page, Voice & Text Input Interface
2	Backend / Logic	3	Speech-to-Text Conversion, NLP Concept Analysis, Mastery Score & Feedback Generation
3	Database / Storage	1	JSON Knowledge Base for concepts, expected answers, and keywords
4	API / Integration	1	Real-Time Flask ↔ NLP Engine Integration with Web Speech API
5	Security / Authentication	0	Not applicable — user authentication is not implemented in the current project scope.